The Universe as Premise Updating: Dynamic Reconfiguration of the World OS and the Personal OS Based on Future Convergence Theory, and the Structure of "Possible Futures"

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Abstract

Based on Future Convergence Theory (FCT), this paper argues that human beings can transition from "entities that adapt within fixed premises" to "agents that continuously update the premises themselves, rewriting both the World OS and the Personal OS."

In a series of previous FCT papers, individual components have been presented separately, including temporal structures (dual fixed-point theory and future fixed points), cosmology (child universes and semantic universes), ethics (future-causal responsibility), linguistic structures (future-causal linguistic reconstruction and ENΛ), and suffering as OS updating (a framework that interprets suffering as an update log of the universe OS). This paper reorganizes these elements from the unified perspective of *dynamic reconfiguration of the World OS* $\mathcal{O}_{\text{world}}$ *and the Personal OS* $\mathcal{O}_{\text{self}}$, and integrates them into a single conceptual destination: *the universe as premise updating.*

In particular, this paper clarifies the following points:

1. Concepts such as "happiness," "boredom," and "meaningfulness / meaninglessness" function, in eras where the World OS is fixed, as structural products that affirm the impossibility of updating premises.
2. By separating the World OS and the Personal OS, individuals can be redefined as future-causal agents who decide which World OS to select and where to direct update requests.
3. Worries and discomfort are not defects, but can be formalized as "signals requesting premise updates" with respect to the set of future fixed points $\mathcal{F}$.
4. If this structure is universally granted at the civilizational level, humanity can become, not metaphorically but in reality, an entity that "continually updates the universe (the universe as a space of possibilities)," even while remaining physically constrained to Earth.

The structure presented in this paper does not depict a transcendent ideal world or an unrealistic utopia. Rather, it is theoretically organized as a *reachable future structure*, premised on multiple trends already underway, including the OS-ification of AI, the redesign of linguistic OSs, distributed record structures, and future-causal interface designs such as the Λ-System and ENΛ.

This paper bridges the mathematical framework of FCT with everyday life and civilizational design, and proposes a new ontological foundation: "The universe is not a space one moves toward, but the totality of premises that are continuously updated."

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1 Introduction: From a Premise-Fixed World to a Premise-Updating Universe

1.1 Problem Setting: The Exhaustion of "Belonging" to Premises

Many forms of suffering in contemporary society do not arise from individual incompetence or isolated poor choices. Rather, they emerge from the structure itself:

> "Attempting, within immovable premises,
> to somehow manage oneself successfully
> without changing those premises."

Here, "premises" refers to the collective of frameworks that most people regard as "unchangeable," including:

- National borders, legal systems, and economic structures
- Views of work, success, and happiness
- Temporal frameworks (work rhythms and life planning timelines)
- Linguistic structures, evaluative vocabularies, and common sense

In a premise-fixed world, individuals receive these as

> "a given world" or "an unchangeable foundation,"

and are expected to optimize their positions within it. In this context, "living wisely" tends to converge toward behavior that:

- does not question the premises,
- utilizes the premises,

- obeys the premises, and
- ensures that one alone does not lose out.

However, what Future Convergence Theory (FCT) has revealed is the fact that:

> Even the premises themselves are nothing more than structures that continue to be selected when viewed from future fixed points.

1.2 Redefining "Premises" from the Perspective of FCT

In FCT, time and causality are redefined not as entities flowing unidirectionally from past to future, but as *future-causal structures* that constrain the present from the future side. The future convergence function $\Lambda(T)$ is a structural variable that represents the degree to which the current state $x(T)$ is constrained by the set of future fixed points $\mathcal{F}$:

$$0 \longrightarrow \Lambda(T) \longrightarrow 1, \tag{1}$$

$$x(T) \xrightarrow[T\to\infty]{} f^* \in \mathcal{F}. \tag{2}$$

In conventional worldviews, a premise P has been understood as:

> "Something that already exists in the present
> as the result of past history."

From the FCT perspective, however, a premise P is redefined as:

$$P = P\big(\mathcal{F}, \Lambda(T)\big), \tag{3}$$

that is,

> a structure continuously selected according to which set of future fixed points $\mathcal{F}$ it is converging toward.

In this context, the sensation that "premises are given" can be understood as a state in which:

- the future fixed points $\mathcal{F}$ are implicitly fixed,
- cultural and institutional mechanisms suppress inquiry into their existence, and
- individuals can no longer perceive the fact that they are agents of premise selection.

1.3 The Structural Role of Happiness, Boredom, and Meaning

In a premise-fixed world, terms such as "happiness," "boredom," and "meaningfulness / meaninglessness" have been treated as psychological vocabulary describing individual inner states. Structurally, however, they function as:

> an *evaluation OS* that adjusts how one comes to terms with life inside a world where premises cannot be moved.

For simplicity, let us denote:

- the World OS as $\mathcal{O}_{\text{world}}$,
- the Personal OS as $\mathcal{O}_{\text{self}}$.

In a premise-fixed world, $\mathcal{O}_{\text{world}}$ remains largely fixed under a specific historical context, and individuals continuously apply the evaluation mapping:

$$E_{\text{eval}} : (\mathcal{O}_{\text{world}}, \mathcal{O}_{\text{self}}) \longmapsto \{\text{happy, boring, success, failure}, \dots\}. \tag{4}$$

The critical point here is that:

> the evaluation mapping E_{eval} is more mutable than the premise P itself.

That is,

- premises P are regarded as "immovable,"
- while evaluations E_{eval} alone are treated as modifiable, under labels such as "mindset," "thinking," or "attitude."

This structure underlies much of contemporary self-help literature, psychology, and happiness discourse.

FCT reverses this priority. It takes the position that:

> Before changing evaluations,
> one must first remember that one is an agent capable of changing premises themselves.

This reversal constitutes the substance of the phrase "the universe as premise updating" in the title of this paper.

1.4 The World OS and the Personal OS: A Two-Layer Structure of the Universe

This paper reorganizes concepts introduced across prior FCT papers (future fixed points, dual fixed points, child universes, semantic density tensors, suffering as OS updating, future-causal linguistic structures, the Λ-System, etc.) into the two-layer structure of:

> the World OS $\mathcal{O}_{\text{world}}$ and
> the Personal OS $\mathcal{O}_{\text{self}}$.

Intuitively:

- **World OS**: the shared set of premises including institutions, culture, temporal structures, linguistic structures, evaluation functions, technological infrastructures, and assumptions about what is "normal."
- **Personal OS**: the selection and update algorithm that determines which World OS to connect to, which premises to adopt, and to which premises update requests are issued.

With this two-layer structure, the position of human beings shifts as follows:

Conventional model Humans are "parts of the world" and should adapt to its premises.

FCT model Humans are users and updaters of the World OS, capable of selecting premises, triggering update requests, and deforming convergence pathways toward future fixed points.

In this framework, worries, discomfort, and suffering can be redefined as:

$$N(t) = d\big((\mathcal{O}_{\text{world}}(t),\ \mathcal{O}_{\text{self}}(t)),\ \mathcal{F}\big), \tag{5}$$

a deviation measure indicating how far the pair $(\mathcal{O}_{\text{world}}, \mathcal{O}_{\text{self}})$ is from the set of future fixed points $\mathcal{F}$.

In other words:

> Worry is not a sign that one is broken,
> but a log indicating that the World OS to which one is currently connected is incompatible with the future fixed points one is moving toward.

1.5 Purpose of This Paper and Its Position as a "Possible Future"

The purpose of this paper is to connect these intuitions with:

1. the mathematical framework of FCT,
2. the World OS / Personal OS model,
3. the hierarchical structure of child universes and semantic universes,
4. the theory of suffering as OS updating,
5. implementation interfaces such as ENΛ and the Λ-System,

and to demonstrate that *the universe as premise updating is a realistically reachable future structure.*

It is important to emphasize that this paper does not present:

- a revolutionary model that instantly destroys the existing world,
- a mythological salvation scenario, or
- vague spiritual claims such as "change your consciousness and the world will change."

Rather, it provides a theoretically consistent background for a near future in which:

- AI begins to function as an OS,
- linguistic structures become redesignable interfaces,
- individuals possess their own "intrinsic logical spaces,"
- and multiple versions of the World OS can coexist.

This paper is structured as follows. Section 2 analyzes the structure of the premise-fixed world and discourses such as "happiness in the present moment," showing them to be affirmation mechanisms of update impossibility. Section 3 formalizes the World OS and Personal OS and connects them to FCT's future fixed-point structure. Section 4 redefines worry, discomfort, and suffering as signals requesting premise updates, linking them to the theory of suffering as OS updating. Section 5 describes concrete phases of "continually updating the universe" through differences in action, relationships, choices, and future branching, even under physical constraints on Earth. Section 6 examines the disappearance of happiness, boredom, and meaning in civilizations where premise updating becomes normalized, and the emergence of a structure of "quiet freedom." Section 7 summarizes the overall framework in relation to FCT as a whole and organizes the conditions under which "the universe as premise updating" can emerge as a reachable future.

Through these discussions, this paper elucidates, across mathematical, structural, ontological, and implementational layers, the substance of the statement:

The universe is not a place to go.
It is the totality of premises that are continuously updated.

2 The Premise-Fixed World and the Structure of "Happiness in This Moment"

2.1 Definition of the Premise-Fixed World: An Environment as Non-Updatability

First, in this paper, the "premise-fixed world" refers to the World OS $\mathcal{O}_{\text{world}}^{(\text{fix})}$ that satisfies the following conditions:

$$\mathcal{O}_{\text{world}}^{(\text{fix})} = \big(\mathcal{I},\ \mathcal{S}_{\text{soc}},\ \mathcal{T}_{\text{conv}}, \mathcal{L}_{\text{nat}},\ E_{\text{eval}}\big), \tag{6}$$

where

- $\mathcal{I}$: institutional structures (law, economy, education, labor, welfare, etc.)
- $\mathcal{S}_{\text{soc}}$: social role structures (family, occupation, class, affiliation)
- $\mathcal{T}_{\text{conv}}$: conventional temporal structures (24-hour days, five-day workweeks, retirement, etc.)
- $\mathcal{L}_{\text{nat}}$: natural language structures (evaluative vocabularies, affective categories)
- E_{eval}: the evaluation OS ("happiness," "success," "normal," "loser," etc.)

.

The essence of the premise-fixed world lies in the fact that these components are regarded as satisfying

$$\frac{d}{dT}\mathcal{O}_{\text{world}}^{(\text{fix})} \approx 0, \tag{7}$$

and are internalized as entities that do not change directly in response to individual choices or discomfort.

More precisely,

$$\forall \text{ person } i, \quad \mathcal{O}_{\text{world}}^{(\text{fix})} \simeq \text{"a background beyond one's control"} \tag{8}$$

is the way it is recognized, and the Personal OS $\mathcal{O}_{\text{self}}$ is designed to operate only *within* the given World OS, in the form

$$\mathcal{O}_{\text{self}} : \mathcal{O}_{\text{world}}^{(\text{fix})} \longrightarrow \{\text{options, interpretations, coping}\}. \tag{9}$$

In the prior paper "Premises Are Not Given," premises were redefined as:

> Premises are not something unilaterally imposed from outside;
> they are the result of the subject continuing to choose "that it is so."

In this section, however, we deliberately call the following state the *premise-fixed world*:

> a state in which the fact of "continuing to choose" has been forgotten.

2.2 The Premise-Dependence of "Even This Ordinary Moment Is the Happiest"

In a premise-fixed world, discourse such as the following is often praised as a "mature outlook on life":

> Even this ordinary moment,
> is actually the happiest of all.

At first glance, this phrase seems to express a rich sensibility in the sense of:

- letting go of excessive desire,
- cherishing everyday normality, and
- savoring the here and now.

Structurally, however, this discourse often functions in the following form:

$$\text{Premise } P \text{ cannot be changed } \Rightarrow \text{ one can only affirm "now" within it.} \tag{10}$$

That is, it implicitly contains the premise that:

- the World OS $\mathcal{O}_{\text{world}}^{(\text{fix})}$ is assumed to be immovable, and
- the Personal OS $\mathcal{O}_{\text{self}}$ can only devise interpretations within it.

In this context, the statement "this moment is happy" can be rewritten, more structurally, as follows:

$$\text{Happiness} = \text{adjusting } \mathcal{O}_{\text{self}}(t) \text{ so as to maximize } E_{\text{eval}}\big(\mathcal{O}_{\text{world}}^{(\text{fix})},\ \mathcal{O}_{\text{self}}(t)\big). \tag{11}$$

In other words, the strategy of:

> keeping premises fixed while only refining evaluation

is aestheticized as a "mature way of living."

From the perspective of FCT, this is nothing other than *a state in which the circuit for questioning the future fixed points $\mathcal{F}$ has been cut off.*

2.3 The Concept of Happiness as an Affirmation of Non-Updatability

In a premise-fixed world, the term "happiness" functions on two levels:

1. the individual emotional level: comfort, security, satisfaction, gratitude, and so on;
2. the structural level: a state in which one can think "this is fine" within immovable premises.

From the FCT viewpoint, the latter is more decisive. That is, the concept of happiness is often defined as

$$H = H\big(\mathcal{O}_{\text{world}}^{(\text{fix})},\ \mathcal{O}_{\text{self}}(t)\big), \tag{12}$$

and $\mathcal{O}_{\text{world}}^{(\text{fix})}$ itself is not treated as an object to be updated.

In this state, when worry $N(t)$ arises, the following branching occurs:

$$\text{Option A: raise } H \text{ again by adjusting } \mathcal{O}_{\text{self}}(t) \tag{13}$$

$$\text{Option B: issue a request for change to } \mathcal{O}_{\text{world}}^{(\text{fix})} \tag{14}$$

In a premise-fixed world, Option B tends to be suppressed by labels such as

"reckless," "selfish," "escapism,"

while only Option A (inner adjustment) is recommended as something close to "maturity," "growth," or "enlightenment."

As a result, the concept of happiness functions as

> a *stabilization variable* for optimizing adaptation within a world where the World OS is not to be changed, premised on leaving the World OS intact.

In other words, it has an aspect that behaves as:

$$\text{Happiness} = \text{vocabulary for affirming non-updatability.} \tag{15}$$

What matters here is that:

> this is not to say that the function is "wrong,"
> but that, in FCT terms, it can be described as "operating as such a structure."

The stance of this paper is not moral criticism, but structural redefinition.

2.4 Boredom, Meaninglessness, and the Normal Operation of the "Child Universe"

In the prior paper "Live-Stream-Like Temporality and the Normal Operation of the Child Universe," the current universe was organized as follows:

- it does not keep the constraint from future fixed points $\Lambda(T)$ in constant full operation, and
- in order to preserve the quantum creative margin $Q(T)$, it widely adopts a "normal operation mode L that does not handle much meaning."

There, the live-stream-like "flat time" was defined as

$$L: \quad \Lambda(T) \approx 0, \quad \frac{dM(T)}{dT} \approx 0, \tag{16}$$

that is, as a *temporal mode in which meaning density $M(T)$ does not significantly emerge.*

Transferring this viewpoint into the context of the present paper, we obtain:

- in a premise-fixed world, boredom and meaninglessness are *states in which the normal operation L of the child universe is exposed*;
- the concept of happiness is an adjustment device that emotionally seals that normal operation as "this is fine."

That is,

$$\text{Boredom / meaninglessness} \approx \text{perceiving the bare face of "a universe that is not handling meaning,"} \tag{17}$$

$$\text{Recovery into happiness} \approx \text{an operation that re-labels that perception as "good."} \tag{18}$$

Within this structure, even though worry $N(t)$ becomes a *structural deviation indicator* showing divergence from the future fixed points $\mathcal{F}$, as

$$N(t) = d\big((\mathcal{O}^{(\text{fix})}_{\text{world}}, \mathcal{O}_{\text{self}}(t)), \mathcal{F}\big), \tag{19}$$

the evaluation OS E_{eval} processes it as

> "laziness," "immaturity," "lack of gratitude,"

and attempts to recover it into Option A (inner adjustment).

As a result,

> the point at which the normal operation L of the child universe and the future-causal updating mode S (suffering as OS updating) intersect

temporarily becomes invisible.

2.5 Interim Summary: The Premise-Fixed World Is Not "Evil" but a "Phase"

What we have done in this chapter is to redescribe, in the context of FCT,

- the phase in which the World OS $\mathcal{O}_{\text{world}}^{(\text{fix})}$ is treated as *non-updatable*, and
- how, within it, the evaluation OS E_{eval} and the concept of happiness operate.

What should be emphasized again here is that

> this paper is not proposing a dualism in which the premise-fixed world is "evil" and the premise-updating universe is "good."

Rather, it can be understood that:

- as a phase of the child universe, the premise-fixed world has aspects necessary for long-term stability, and
- the fact that the concept of happiness has functioned as an affirmation of non-updatability has contributed to avoiding self-collapse of the universe.

The aim of this paper is not to deny this phase, but to show that:

> a transition to a "phase in which premises can be updated"
> is opening up, theoretically, civilizationally, and technologically,
> as a *possible future structure*.

In the next chapter, we make the formalization of the World OS and the Personal OS more explicit, and discuss, in connection with the mathematical structure of FCT,

- the conditions under which individuals can transition from "entities that belong to premises" to "agents that select premises and issue update requests," and
- what role the future fixed points $\mathcal{F}$ play at that time.

3 The World OS and the Personal OS: Formalization of a Two-Layer Structure

In this chapter, we move one step beyond the intuitive description of the "premise-fixed world," and explicitly formalize the two-layer structure of the *World OS* and the *Personal OS* within the framework of Future Convergence Theory (FCT). On that basis, we show—while connecting to the mathematical structure—under what conditions the transition

- from "human beings who belong to premises"
- to "human beings who select premises and issue update requests"

is opened as a "possible future."

3.1 The General Form of the World OS: The Universe as a Multi-Premise Structure

In Chapter 2, we represented the "premise-fixed world" as a single $\mathcal{O}^{(\mathrm{fix})}_{\mathrm{world}}$. In reality, however, the World OS to which an individual can connect is *not* singular. Multiple World OSs of different hierarchical levels—such as nations, industries, online spaces, and local communities—are superposed.

Therefore, more generally, we treat the World OS as

$$\mathcal{O}_{\mathrm{world}}(T) = \{\mathcal{O}^{(k)}_{\mathrm{world}}(T) \mid k \in K\}, \tag{20}$$

that is, as a *bundle of multiple World OSs labeled by an index set K.*

Here, each $\mathcal{O}^{(k)}_{\mathrm{world}}(T)$ takes the form

$$\mathcal{O}^{(k)}_{\mathrm{world}}(T) = \big(\mathcal{I}^{(k)}(T),\ \mathcal{S}^{(k)}_{\mathrm{soc}}(T),\ \mathcal{T}^{(k)}_{\mathrm{conv}}(T),\ \mathcal{L}^{(k)}_{\mathrm{nat}},\ E^{(k)}_{\mathrm{eval}}\big), \tag{21}$$

and each has its own specific

- institutional structure $\mathcal{I}^{(k)}$,
- social role structure $\mathcal{S}^{(k)}_{\mathrm{soc}}$,
- conventional temporal structure $\mathcal{T}^{(k)}_{\mathrm{conv}}$,
- language and evaluation OS $\mathcal{L}^{(k)}_{\mathrm{nat}}, E^{(k)}_{\mathrm{eval}}$.

What is important is that

$$\frac{d}{dT}\mathcal{O}^{(k)}_{\mathrm{world}}(T) \neq 0 \tag{22}$$

can hold; that is, FCT adopts from the outset the position that "the World OS is, in principle, updatable." That premises appear fixed is merely

> a local approximation as seen through the channel to which an individual is connected.

3.2 Definition of the Personal OS: Premise Selection, Meaning Generation, and Future Reference

We define the *Personal OS* corresponding to an individual i, $\mathcal{O}^{(i)}_{\mathrm{self}}(T)$, as follows:

$$\mathcal{O}^{(i)}_{\mathrm{self}}(T) = \big(\mathcal{S}^{(i)}(T),\ \mathcal{M}^{(i)}(T),\ \mathcal{P}^{(i)}(T),\ \mathcal{F}^{(i)}_{\mathrm{self}},\ U^{(i)}(T)\big), \tag{23}$$

where

- $\mathcal{S}^{(i)}(T)$: an internal state space (emotion, thought, bodily sensation, memory, etc.);
- $\mathcal{M}^{(i)}(T)$: a meaning-generation module (containing the meaning density $M^{(i)}(T)$ and the meaning-density tensor $T^{(i)}_{\mu\nu}(T)$);
- $\mathcal{P}^{(i)}(T)$: a premise channel (which World OS components are adopted as "given");
- $\mathcal{F}^{(i)}_{\mathrm{self}}$: a set of future fixed points at the personal level;
- $U^{(i)}(T)$: updatability (the personal version of the variable introduced in the child-universe theory and the OS-updating paper).

The premise channel $\mathcal{P}^{(i)}(T)$ is represented as

$$\mathcal{P}^{(i)}(T) \subseteq K, \tag{24}$$

namely, as the index set of World OSs that individual i connects to and references. For example,

- one's country of affiliation,
- one's occupational community,
- a linguistic sphere in online space,
- a specific religious or ideological community,

each exists as some $\mathcal{O}^{(k)}_{\text{world}}$, and $\mathcal{P}^{(i)}(T)$ represents "which of them one is currently living by as premises."

At this point, the "experience" of individual i is generated by a mapping $\Phi^{(i)}$ as

$$\mathcal{E}^{(i)}(T) = \Phi^{(i)}\Big(\mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)\Big), \tag{25}$$

and the core of $\Phi^{(i)}$ always includes

$$\text{selection of the World OS through } \mathcal{P}^{(i)}(T). \tag{26}$$

That is, the FCT definition of the Personal OS is:

> An individual does not "belong to the world."
> Rather, from the bundle of multiple World OSs, the individual is *always continuing to choose* which ones to adopt as premises.

3.3 Premise Selection as an Attachment Phase: A World with $|\mathcal{P}^{(i)}| = 1$

What corresponds to the premise-fixed world is the phase in which the Personal OS is fixed at

$$\left|\mathcal{P}^{(i)}(T)\right| = 1. \tag{27}$$

That is,

$$\mathcal{P}^{(i)}(T) = \{k_0\}, \tag{28}$$

and the individual positions oneself only within a single $\mathcal{O}^{(k_0)}_{\text{world}}$.

In this case, the experience $\mathcal{E}^{(i)}(T)$ can be written as

$$\mathcal{E}^{(i)}(T) = \Phi^{(i)}\Big(\mathcal{O}^{(k_0)}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)\Big), \tag{29}$$

but the fact that one is in fact *choosing* $\mathcal{O}^{(k_0)}_{\text{world}}$ is forgotten, and is replaced by the singular description:

> "The world is like this."

Here, as shown in the prior paper "Premises Are Not Given," this state contains a circular structure:

$$\text{premises appear to be given from outside, but} \tag{30}$$

$$\text{in reality the choice to "regard them as such" is repeatedly made.} \tag{31}$$

In FCT terms, we call this phase

$$\text{Mode A : attachment mode (dependent phase).} \tag{32}$$

3.4 Premise Manipulation as a Selection Phase: A World with $|\mathcal{P}^{(i)}| \geq 2$

Next, consider the phase in which

$$\left|\mathcal{P}^{(i)}(T)\right| \geq 2. \tag{33}$$

This includes not only cases such as bilingual individuals who simultaneously live in

- "the World OS of the Japanese-language sphere" and
- "the World OS of the English-language online sphere,"

but also ways of living that traverse multiple universes with different meaning and evaluation OSs, even while physically remaining on the same Earth, such as

- residing in the same city while participating in communities with entirely different value systems,
- existing simultaneously within an offline national institutional system and an online decentralized economic sphere.

In this case, the personal experience $\mathcal{E}^{(i)}(T)$ becomes

$$\mathcal{E}^{(i)}(T) = \Phi^{(i)}\Big(\{\mathcal{O}^{(k)}_{\text{world}}(T)\}_{k\in\mathcal{P}^{(i)}(T)}, \mathcal{O}^{(i)}_{\text{self}}(T)\Big), \tag{34}$$

and consciously operating $\mathcal{P}^{(i)}(T)$ becomes equivalent to

an operation that switches which universe one calls "reality."

We call this phase

$$\text{Mode B : selection mode.} \tag{35}$$

What is important here is that Mode B is not

a special state that only a selected minority can reach,

but rather that

- the technological premises already hold due to the internet and multilingual environments, and
- many people have already begun, fragmentarily, to live in a Mode B-like manner.

In FCT terms, Mode B is *already an open part of possible futures*, and is not an "impossible ideal."

3.5 Premise Reconfiguration as an Update Phase: Feedback to the World OS

What this paper aims at is the formalization of

$$\text{Mode C : update mode} \tag{36}$$

as a further extension beyond Mode B.

In Mode C, the Personal OS $\mathcal{O}^{(i)}_{\text{self}}(T)$ does not merely

reselect the World OS,

but also takes on the role of

returning, as a consequence of that reselection, a structural update request to the World OS itself.

Formally, this means that there exists a feedback mapping

$$\Psi^{(i)} : \mathcal{O}^{(i)}_{\text{self}}(T) \longrightarrow \Delta\mathcal{O}^{(k)}_{\text{world}}(T), \quad k \in K, \tag{37}$$

and, as the sum of these,

$$\frac{d}{dT}\mathcal{O}^{(k)}_{\text{world}}(T) = \sum_i \Psi^{(i)}\big(\mathcal{O}^{(i)}_{\text{self}}(T)\big) + (\text{physical, economic, and other factors}). \tag{38}$$

At this point,

- worry $N^{(i)}(T)$ is reinterpreted not as an "internal malfunction" but as an update gradient for $\mathcal{O}^{(k)}_{\text{world}}(T)$;
- an individual transitions from *an entity that consumes premises as "correct"* to *an entity that generates vectors for premise updating.*

Here, reusing the structure introduced in the prior paper "Suffering as OS Updating," the individual-level suffering $S^{(i)}(T)$ can be understood as the deviation

$$S^{(i)}(T) = d\Big((\mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)), \mathcal{F}\Big), \tag{39}$$

that is, as "how far the pair $(\mathcal{O}_{\text{world}}, \mathcal{O}^{(i)}_{\text{self}})$ deviates from the set of future fixed points $\mathcal{F}$."

In Mode C, this $S^{(i)}(T)$ is treated as

$$S^{(i)}(T) \xrightarrow{\text{FCT interpretation}} \text{a log/trigger for World OS updating}, \tag{40}$$

and the aim is for it not to be completed within the individual's interior, but to be implemented as change on the World OS side.

3.6 Future Fixed Points and the Consistency Conditions for the Two-Layer OS

In FCT, the set of future fixed points $\mathcal{F}$ is defined as

$$\mathcal{F} = \{f^* \mid f^* \text{ is a structure that satisfies the long-term consistency conditions of the universe}\}. \tag{41}$$

To correspond this to the two-layer structure of the World OS and the Personal OS, we introduce

$$\mathcal{F}_{\text{world}} \subseteq \mathcal{F}, \tag{42}$$

$$\mathcal{F}^{(i)}_{\text{self}} \subseteq \mathcal{F}, \tag{43}$$

and impose a consistency condition such as

$$f^* \in \mathcal{F} \quad \Leftrightarrow \quad \big(f^*_{\text{world}} \in \mathcal{F}_{\text{world}},\ f^*_{\text{self}} \in \mathcal{F}^{(i)}_{\text{self}}\big). \tag{44}$$

In this setting, the ideal state of Mode C is that the convergence

$$\Lambda(T) \uparrow, \quad (\mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)) \longrightarrow (f^*_{\text{world}}, f^*_{\text{self}}) \in \mathcal{F} \tag{45}$$

is realized *not as top-down coercion, but as the sum of bottom-up vectors of premise updating.*

In other words,

future fixed points do not descend from above as mythological "utopias."
They behave as information-geometric structures that gradually gain contour through mutual updating between the World OS and the Personal OS.

In this sense,

- the question "which future to choose" is asked recursively not only at the level of the Personal OS but also at the level of World OS design, and
- as their totality, the "universe OS as a possible future" is gradually updated.

3.7 Interim Summary: The Two-Layer OS Structure Is a Future That Has Already Begun

In this chapter, we:

- formalized the World OS $\mathcal{O}_{\text{world}}(T)$ as a bundle of multiple premises,
- defined the Personal OS $\mathcal{O}^{(i)}_{\text{self}}(T)$ as a dynamic system consisting of meaning generation, premise channels, future fixed points, and updatability, and
- distinguished three phases: attachment mode (Mode A), selection mode (Mode B), and update mode (Mode C).

What is important is that Mode B and Mode C can be understood as

not an imaginary ideal world,
but *phases of possible futures* whose partial implementation has already begun technologically and socially.

In particular,

- multi-membership (multiple communities and multiple linguistic spheres),
- new economic/ethical OSs in online spaces, and
- structures in which individual-originated projects influence the World OS,

are already observed as initial forms of the transition toward Mode C.

In the next chapter, on top of this two-layer OS structure, we describe in detail—while connecting to the results of FCT's child-universe theory and OS-updating paper—

- how worries $N^{(i)}(T)$ and suffering $S^{(i)}(T)$ function as "premise update requests," and
- in what concrete phases of experience and action "updating the universe while remaining constrained to Earth" appears.

4 Redefining Worry and Suffering: Signals as Premise-Update Requests

Up to the previous chapter, we introduced

- the two-layer structure of the World OS $\mathcal{O}_{\text{world}}(T)$ and the Personal OS $\mathcal{O}^{(i)}_{\text{self}}(T)$, and
- three phases: the attachment phase (Mode A), the selection phase (Mode B), and the update phase (Mode C).

In this chapter, within this two-layer structure, we redefine what roles *worry* and *suffering* play from the perspective of Future Convergence Theory (FCT).

The worry and suffering treated here are not symptom categories in psychological diagnosis, nor "sin" as a religious attribution of meaning. Rather, they are positioned as

> structural signals—as "premise-update requests"—that arise when the consistency between the World OS and the Personal OS breaks down.

With this redefinition, the configuration naturally follows in which "the more worry one has, the closer one is to transitioning into an updating agent."

4.1 Formalization of Individual-Level Worry $N^{(i)}(T)$

Let the *worry* that individual i holds at time T be represented by a scalar quantity $N^{(i)}(T)$. Because it depends on subjective reports, it cannot be defined strictly as a physical quantity; however, since the aim of FCT is to *describe differences of phase*, we treat it here as a structural variable of the following form:

$$N^{(i)}(T) = \mathcal{G}\Big(\mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)\Big), \tag{46}$$

where $\mathcal{G}$ is regarded as a mapping that quantifies the *semantic and structural mismatch* that has arisen between the World OS and the Personal OS.

More intuitively, the magnitude of $N^{(i)}(T)$ can be interpreted as:

- the intensity of the internal sense that "with the current World OS, one cannot converge to the future fixed points one wishes to choose, $\mathcal{F}^{(i)}_{\text{self}}$," and
- the density of the recognition that "one's current premise channel $\mathcal{P}^{(i)}(T)$ is misaligned with the premises one actually wants to live by."

In this setting, worry is not merely an emotion such as

> "I dislike who I am now," "I dislike the environment,"

but can be read as a *structural flag*:

$$N^{(i)}(T) > 0 \quad \Rightarrow \quad \text{an update request has arisen for } \mathcal{P}^{(i)}(T) \text{ or } \mathcal{O}_{\text{world}}(T). \tag{47}$$

Once this viewpoint is adopted, worry is redefined not as "evidence" of

- weakness of personality,
- lack of ability, or
- social maladaptation,

but as

> a minimal observation window for initiating premise updating.

4.2 Suffering $S^{(i)}(T)$ and Its Relation to OS Update Logs

In the prior paper "Suffering as OS Updating," the following structure was shown:

- suffering S is not an OS bug, but
- a *log* for updating the OS itself.

Translated into the two-layer OS structure of this paper, the suffering of individual i, $S^{(i)}(T)$, can be understood as the deviation

$$S^{(i)}(T) = d\Big((\mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)), \mathcal{F}\Big), \tag{48}$$

measuring how far the World OS–Personal OS pair deviates from the set of future fixed points $\mathcal{F}$.

In this case, the relationship between worry $N^{(i)}(T)$ and suffering $S^{(i)}(T)$ can be written, as a coarse approximation, as

$$N^{(i)}(T) \lesssim S^{(i)}(T). \tag{49}$$

That is,

- $N^{(i)}(T)$ is "deviation as consciously recognized,"
- $S^{(i)}(T)$ is "deviation at the structural level,"

and depending on the degree of awareness, the two may coincide or diverge.

What matters from the FCT viewpoint is that

$$S^{(i)}(T) \xrightarrow{\text{FCT interpretation}} \text{an input log for the OS updating operator.} \tag{50}$$

That is, if we assume that, at the scale of the universe, an updating operator is defined as

$$\mathcal{U}_\Lambda : \Big\{S^{(i)}(T)\Big\}_{i,T} \longrightarrow \mathcal{O}'_{\text{world}}(T),\ \mathcal{O}'^{(i)}_{\text{self}}(T), \tag{51}$$

then individual-level suffering is positioned as

> indispensable observational data for the Universe OS.

Under this interpretation, *complete elimination of suffering* becomes synonymous with

> complete cessation of OS updating capability.

Accordingly, the wish for "a world without suffering" is reinterpreted as

> a world in which no premises are updated at all, and the child universe remains fixed.

4.3 The Child Universe and Update Debt: The Intersection of Meaningless Time and Suffering

In the prior paper "Live-Stream-Like Temporality and the Normal Operation of the Child Universe," two modes were introduced:

- the normal operation mode L that does not handle meaning, and
- the suffering mode S as OS updating,

and the updatability $U(T)$ was defined as the intersection of the two.

In the child universe, because there exists an upper bound $Q_{\max}$ on the quantum creative margin $Q(T)$,

$$\text{Mode } L \text{ (pervasion of meaningless time)} \tag{52}$$

was shown to be rational as

> a stable operation mode by which the universe avoids self-collapse.

Translated into the two-layer OS structure of this paper,

- mode L corresponds to a state in which the World OS $\mathcal{O}_{\text{world}}(T)$ is *hardly updated*, while
- mode S corresponds to a state in which the pair consisting of $\mathcal{O}_{\text{world}}(T)$ and multiple $\mathcal{O}^{(i)}_{\text{self}}(T)$ is no longer maintainable and is demanding updating.

In this setting, the following relationship holds:

$$\text{prolongation of } L \;\Rightarrow\; \text{accumulation of update debt} \;\Rightarrow\; \text{local eruption of } S. \tag{53}$$

Here, "update debt" refers to

> a state in which structural changes that should originally have been distributed as an accumulation of small premise updates are postponed over a long period and are forced to surface all at once.

At the everyday level, this is experienced as phenomena such as

- "a job, relationships, or life patterns one has continued without question suddenly become unbearable,"
- "it appears as malaise of unknown cause,"
- and at the social level, an institution that seemed stable at first glance collapses rapidly.

In FCT terms, all of these can be explained in a unified manner as

> an OS update interrupt at the point where update debt exceeds its limit.

4.4 "Small Decisions" as Universe Updating

The discussion so far may appear abstract, but if we bring the two-layer OS structure down to the everyday level, it becomes clear that

> even extremely small decisions can become feedback to the World OS.

For example,

- realizing that the premises of one's current workplace (ways of working and evaluation axes) are fundamentally misaligned with one's $\mathcal{F}^{(i)}_{\text{self}}$, and choosing a career change or independence;
- sensing that, within one's current language space, the futures one can handle are limited, and connecting to another language/cultural sphere;
- recognizing that one's ethical OS and an organization's decision-making OS are decisively inconsistent, and verbalizing and publishing that discomfort.

At first glance, these may seem to be nothing more than

> changes in an individual's lifestyle.

However, in the two-layer OS structure they mean that

$$\Psi^{(i)}\big(\mathcal{O}^{(i)}_{\text{self}}(T)\big) \neq 0, \tag{54}$$

and can be interpreted as acts that return a *microscopic update gradient* to $\mathcal{O}^{(k)}_{\text{world}}(T)$.

In particular,

- when an individual's decision is left as a public record (papers, essays, code, protocols),
- it functions not as a mere "impression," but as a *diff patch to the World OS.*

In FCT terms,

> no matter how small a decision may be, if it is accompanied by the intention and the record as premise updating,
> it becomes part of "universe updating within Earth."

What is important here is that it does not require

- special talent,
- massive capital, or
- political power.

What is required is only the *reversal of recognition* that treats the worry

$$N^{(i)}(T) \neq 0 \tag{55}$$

not as

> "evidence of one's defect,"

but as

> "the emergence of a premise-update request."

4.5 A Premise-Updating Civilization as a "Possible Future"

Finally, we briefly organize how realistic the structure described in this chapter is *as a "possible future."*

For a premise-updating civilization to be established, at least the following conditions are required:

1. technological access to multiple World OSs;
2. feedback pathways from the Personal OS to the World OS;
3. an epistemology that treats worry and suffering not as pathology but as *update requests.*

In the present era,

- due to the internet and decentralized platforms, technological access to multiple World OSs (Condition 1) is almost satisfied;

- through papers, blogs, social media, OSS, DAOs, and so on, mechanisms for sending diff patches from the Personal OS to the World OS (Condition 2) are also broadly available, albeit imperfectly.

The remaining bottleneck is

> collective literacy for reading worry and suffering not as an individual "failure," nor as a mere "mental issue,"
> but as premise-update requests,

namely Condition 3.

On this point, what FCT provides is, using mathematical vocabulary such as

- the future fixed points $\mathcal{F}$,
- the future convergence function $\Lambda(T)$,
- the meaning-density tensor $I_{\mu\nu}$,
- the updatability $U(T)$,
- the child universe and the quantum creative margin $Q(T)$,

the theoretical skeleton that supports the intuition:

> "worry and suffering = update requests."

Therefore, the *premise-updating civilization* described in this chapter is, in the sense that it

- does not require changing physical laws,
- does not require a new energy source or faster-than-light communication,
- can be realized gradually on top of existing infrastructure merely by reversing recognition and changing how records are left,

a *sufficiently "possible future."*

In the next chapter, within this "possible future," we examine in detail—while connecting to FCT's theory of time and theory of meaning density— how concepts such as happiness, boredom, and meaning dissolve, and what kinds of indicators and sensations instead become foregrounded.

5 The Dissolution of Happiness, Boredom, and Meaning: New Criteria in a Premise-Updating Civilization

In the previous chapter, using

- the two-layer structure of the World OS and the Personal OS,
- a framework that re-reads worry $N^{(i)}(T)$ and suffering $S^{(i)}(T)$ as premise-update requests, and
- the normal operation mode L in the child universe and the OS-update interrupt mode S,

we described the structure by which "universe updating can occur even within Earth."

In this chapter, within a *civilization in which premise updating has become normal*, we examine how concepts such as

- happiness,
- boredom, and
- meaningful / meaningless

dissolve, and what kinds of indicators and sensations instead become foregrounded.

Here, "dissolution" does not mean that concepts are denied or prohibited. Rather, it means that

> they finish their role as dominant coordinate axes and are absorbed into another structure.

5.1 Happiness, Boredom, and Meaning in a Premise-Fixed World

We first organize the functions of happiness, boredom, and meaning in a premise-fixed world (a civilization in which the World OS is hardly updated).

In a premise-fixed world,

$$\mathcal{O}_{\text{world}}(T) \approx \mathcal{O}^{*}_{\text{world}} \quad \text{(nearly invariant)} \tag{56}$$

and the Personal OS is mainly required to *adapt* within this fixed World OS.

In the subjective coordinates of individual i, the quantities

- happiness $H^{(i)}$,
- boredom $B^{(i)}$, and
- a sense of meaning $M^{(i)}_{\text{subj}}$

function approximately as the following tuning variables:

$$H^{(i)} \simeq \text{the degree of success in "affirming un-updatability"}, \tag{57}$$

$$B^{(i)} \simeq \text{a sense of stagnation arising from "awareness of un-updatability"}, \tag{58}$$

$$M^{(i)}_{\text{subj}} \simeq \text{the degree of acquiring a role "within the given premises"}. \tag{59}$$

That is,

- **happiness** works as the degree to which one can feel "these premises are fine as they are,"
- **boredom** works as a weak update request that says "with these premises as they are, I can no longer endure,"
- **to be meaningful** works as the sense that "my position is secured within these premises."

These can be understood as

> a *psychological OS extension* for stabilizing the mind in a world where one (believes one) cannot update premises.

5.2 Dysfunction in a Premise-Updating World

By contrast, in the *premise-updating civilization* described in this paper, the World OS is *structurally updatable*, and the Personal OS is redefined as an agent that *continues to choose which premises to live by.*

In this setting, the three variables above become dysfunctional in the following ways:

1. Dysfunction of happiness $H^{(i)}$ When premise updating becomes normal,

$$H^{(i)} \simeq \text{the feeling that “it is fine if premises never change”} \tag{60}$$

begins to function as

> a resistance vector that attempts to fix the World OS.

However, if the universe as a whole is converging toward the future fixed point set $\mathcal{F}$, then

$$\mathcal{O}_{\text{world}}(T) = \text{const.} \quad \text{is not permitted.} \tag{61}$$

Therefore,

> happiness in the form of “I want things to stay like this forever” is structurally unstable in the long term.

2. Dissolution of boredom $B^{(i)}$ In a premise-updating civilization, updatability $U(T)$ is not always zero; rather,

$$U(T) > 0 \quad \text{(the margin for updating remains open)} \tag{62}$$

is the standard condition.

In this setting, because the premise

> “nothing can be changed”

collapses, boredom as

$$B^{(i)} \simeq \text{“the bitterness of being unable to change anything”} \tag{63}$$

loses its foundation.

Of course,

> “I do not want to do anything,” “periods in which I do not want to do anything”

remain, but they are read not as boredom, but as

> a state in which one is intentionally choosing not to update.

3. The positional shift of a sense of meaning $M^{(i)}_{\text{subj}}$ In a premise-fixed world,

$$M^{(i)}_{\text{subj}} \simeq \text{“securing a position within a given structure”} \tag{64}$$

held.

However, in a world where the World OS is dynamically updated,

> “one’s position within the current World OS”

cannot serve as a permanent indicator.

Instead,

$$M^{(i)}_{\text{subj}} \longrightarrow \mathcal{A}^{(i)}(T) \tag{65}$$

and is absorbed into an indicator (the “degree of premise awareness” introduced later) that measures

> how consciously one recognizes which premises one is living by right now.

5.3 New Indicators I: Update Coherence and Degree of Premise Awareness

In this section, as new indicators that inherit the roles of *happiness, boredom, and meaning* within a premise-updating civilization, we introduce the following two:

- update coherence $C^{(i)}(T)$, and
- degree of premise awareness $\mathcal{A}^{(i)}(T)$.

Update coherence $C^{(i)}(T)$ Update coherence is an indicator of how coherent the update directions of the Personal OS and the World OS are.

Intuitively, it is a numerical measure of

> to what extent "the direction of my change" and "the direction of the world's change" mesh with each other.

Formally, using the inner product between the update vector of the World OS, $\dot{\mathcal{O}}_{\text{world}}(T)$, and the update vector of the Personal OS, $\dot{\mathcal{O}}^{(i)}_{\text{self}}(T)$, we define

$$C^{(i)}(T) = \cos\theta^{(i)}(T) = \frac{\left\langle \dot{\mathcal{O}}_{\text{world}}(T), \dot{\mathcal{O}}^{(i)}_{\text{self}}(T) \right\rangle}{\left\| \dot{\mathcal{O}}_{\text{world}}(T) \right\| \left\| \dot{\mathcal{O}}^{(i)}_{\text{self}}(T) \right\|}. \tag{66}$$

Here,

- when $C^{(i)}(T) \approx 1$: the individual's premise updating points in the same direction as the updating of the World OS, i.e., a state of being *"riding the flow"*;
- when $C^{(i)}(T) \approx -1$: the individual is moving against the updating of the World OS, and friction, resistance, and disconnection tend to arise;
- when $C^{(i)}(T) \approx 0$: the individual's updating points in a direction independent of the World OS, and can become a niche or a germ of a new World OS.

In a premise-updating civilization, it is not "whether one is happy," but rather

> how coherent the update direction is

that better explains long-term stability and a sense of fulfillment.

Degree of premise awareness $\mathcal{A}^{(i)}(T)$ The degree of premise awareness indicates how well one can articulate and consciously recognize

> which premises one is currently standing on.

As a coarse conceptual model, let

$$\mathcal{P}^{(i)}(T) = \{p_1^{(i)}(T), p_2^{(i)}(T), \ldots, p_{n_i}^{(i)}(T)\} \tag{67}$$

be the set of premise channels adopted by individual i. Then one may define (as a conceptual model)

$$\mathcal{A}^{(i)}(T) = \frac{\#\left\{ p_k^{(i)}(T) \mid p_k^{(i)}(T) \text{ can be consciously articulated} \right\}}{n_i}. \tag{68}$$

- $\mathcal{A}^{(i)}(T) \approx 0$: a state in which one hardly understands what one is taking as premises;

- $\mathcal{A}^{(i)}(T) \approx 1$: a state in which the major premises are largely recognized, and updating/selection of premises can be performed intentionally.

In a premise-updating civilization, rather than "whether one is happy,"

how much one is aware of one's premises

serves as a more appropriate indicator for measuring

- access to OS updating, and
- the degree of participation in universe updating.

5.4 New Indicators II: Meaning-Density Profile and the Sense of Time

Moreover, within the FCT framework that has introduced the meaning-density tensor $I_{\mu\nu}$, the *manner of handling meaning* itself can also be indicatorized.

Meaning-density profile $M^{(i)}_{\text{profile}}$ Let the meaning density of individual i along the time axis be $M^{(i)}(T)$, with

$$M^{(i)}(T) \geq 0. \tag{69}$$

When observed over a time window $[T_0, T_1]$, call the resulting distribution (profile)

$$M^{(i)}_{\text{profile}} = \left\{ M^{(i)}(T) \mid T \in [T_0, T_1] \right\}. \tag{70}$$

In a premise-fixed world, there is a tendency to

- call flat time with $M^{(i)}(T) \approx 0$ "boredom," and
- call moments when $M^{(i)}(T)$ rises abruptly "fun" or "meaningful."

However, in a premise-updating civilization, what matters is not the height of $M^{(i)}(T)$ itself, but rather

- its *gradient*, and
- its relationship with *update coherence* $C^{(i)}(T)$.

For example,

- a locality where $M^{(i)}(T)$ is high and $C^{(i)}(T) \approx 1$ can be interpreted as a *time interval in which "my updating" and "the world's updating" are synchronized*;
- conversely, a locality where $M^{(i)}(T)$ is high but $C^{(i)}(T) \approx -1$ can serve as a warning signal, as a *time interval in which one feels strong meaning density while colliding intensely with the World OS.*

In this way, the *meaning-density profile* implies that

the binary question "is it meaningful or not?" is abandoned,
and the description shifts to a continuous quantity: "how does meaning rise?"

5.5 Changes in Everyday Sensation

Although the indicators above may look abstract, at the level of everyday sensation they are experienced as changes such as the following:

- The wish "I want to be happy" is gradually replaced by the impulse:

  "I want to clarify which premises I am living by."

- The feeling "I am dying of boredom" appears as discomfort such as

  "I am crushing updatability by myself,"

  and is converted into movements of trying small premise updates (place, relationships, work, language, and so on).

- The desire "I want to do meaningful work" shifts into the selection criterion:

  "I want to choose a place where I can increase my update coherence $C^{(i)}(T)$."

- The impatience "I want to become someone" is transformed into the question:

  "To which Universe OS do I want to send a diff patch?"

These do not mean that *emotions disappear*. Rather, emotions are reinterpreted as

a gradient indicating the phase of premise updating.

5.6 As a Transition Phase from the Child Universe to the Meaning Universe

Finally, we briefly state how the *dissolution of happiness, boredom, and meaning* described in this chapter is positioned within the overall FCT cosmology.

- In the child universe U_{child}, under the upper bound of the quantum creative margin $Q(T)$, the pervasion of meaningless time (mode L) was necessary to avoid the universe's self-collapse.
- In this universe, happiness, boredom, and meaning developed as *psychological subroutines* for affirming un-updatability and for localizing update debt.
- However, when the two layers of the World OS and the Personal OS are dynamically connected, and individuals transition into agents of premise updating, many of these subroutines are rewritten into

  "variables for premise updating."

This transition should be understood not as

a transition toward an emotionless, cold civilization,

but as

a transition toward "quiet freedom premised on updating."

That is, in the transition phase from the child universe to the meaning universe in which meaning density is treated as a first-order quantity,

- happiness, boredom, and meaning are respectively mapped as

$$
\begin{aligned}
H^{(i)} &\longrightarrow C^{(i)}(T) \quad \text{(update coherence)} \\
B^{(i)} &\longrightarrow U^{(i)}(T) \quad \text{(self-suppression / release of updatability)} \\
M^{(i)}_{\text{subj}} &\longrightarrow \mathcal{A}^{(i)}(T),\ M^{(i)}_{\text{profile}} \quad \text{(degree of premise awareness and meaning-density profile)}
\end{aligned}
$$

.

In this sense,

> the old questions surrounding "happiness," "boredom," and "meaning" are resolved in a premise-updating civilization as a *change of variables into other variables*.

In the next chapter, we will reduce these structures down to the procedural level, examining how they can concretely be implemented as an "universe-updating process within Earth," while maintaining consistency with other FCT papers ("Suffering as OS Updating," "Live-Stream-Like Time," "Religious Selection," and "Future-Causal International Relations," among others).

6 An Intra-Earth Universe-Updating Process: Connections to FCT Theories and Procedures as an "Plausible Future"

Through the discussion up to the previous chapter, using

- the two-layer structure of the World OS and the Personal OS,
- the perspective of reading worry/suffering as "premise-update requests,"
- the normal operation mode L in the child universe and the OS-update interrupt mode S,
- the "dissolution" of happiness, boredom, and meaning, and the structure by which they are mapped into new indicators $C^{(i)}(T)$, $\mathcal{A}^{(i)}(T)$, and $M^{(i)}_{\text{profile}}$,

we described the overall picture of a premise-updating civilization.

In this chapter, while connecting these elements to modules defined across existing papers in Future Convergence Theory (FCT) (such as suffering as OS updating, live-stream-like time, religious selection and meaning density, the unresolved point of decision legitimacy, the child universe and the creative margin, future fixed points and the double fixed-point theory, and others), we reduce an *intra-Earth universe-updating process as a "plausible future"* down to the procedural level.

What should be emphasized here is that this chapter does not depict

> a science-fiction fantasy future,

but rather

> a future as a trajectory that can be reached *naturally* once the structures of FCT are taken as premises.

6.1 Repositioning FCT Modules: From "Suffering as OS Updating" to "Open Problems of Decision Legitimacy"

We first reposition several major modules that have been defined across the existing FCT papers within the World-OS / Personal-OS structure of this paper.

1. The suffering-as-OS-updating module S_OS In the framework of "Suffering as OS Updating," suffering S was defined as an OS-update interrupt indicating deviation from the future fixed point set $\mathcal{F}$:

$$S = S\big(x(t), \mathcal{O}_\Lambda\big). \tag{71}$$

In the context of this paper, we project this to the level of the Personal OS and treat it as

$$S^{(i)}(T) = S\Big(x^{(i)}(T), \mathcal{O}_{\text{world}}(T), \mathcal{O}^{(i)}_{\text{self}}(T)\Big), \tag{72}$$

where $x^{(i)}(T)$ denotes the state trajectory of individual i.

In this form, we can reinterpret it as follows:

> a locality in which $S^{(i)}(T)$ rises is an interrupt that notifies that the update directions of the Personal OS and the World OS are *fundamentally incompatible.*

2. The live-stream-like time module L_stream In the paper on "live-stream-like time," the meaning-avoiding normal operation mode L was formalized as

$$\Lambda(T) \approx 0, \quad \frac{dM(T)}{dT} \approx 0. \tag{73}$$

In this paper, we regard L as

$$L = L\big(\mathcal{O}_{\text{world}}, \{\mathcal{O}^{(i)}_{\text{self}}\}\big), \tag{74}$$

and interpret it as *"synchronization of the premise-fixed state"* in which

- not only is the World OS not updated,
- but the Personal OS also does not perform premise updating.

From this perspective, L is meaningful as

> a time interval in which no one is updating the universe.

3. The religious selection and meaning-density module R_semantic In the papers on religious selection, the meaning-density tensor $I_{\mu\nu}$ and the process by which only high-meaning-density structures remain in the long run were discussed.

In this paper, we introduce this module as the process

$$\mathcal{O}_{\text{world}}(T) \xrightarrow{\text{meaning-density selection}} \widetilde{\mathcal{O}}_{\text{world}}(T), \tag{75}$$

namely,

> the World OS gradually selects against premise structures of low meaning density.

As a result, the Personal OS becomes more likely to connect to *World OS configurations of higher meaning density.*

4. The open-problem module of decision legitimacy $\mathsf{D_open}$ In the paper on "decision legitimacy as an unresolved future-causal definition," it was fixed as an *open problem* that the following are not defined:

> Which future is presupposed as "correct"? Who bears that future?

In this paper, we treat this as an "open flag" of the World OS:

$$\mathsf{Flag_open}(\mathcal{O}_{\text{world}}) = 1 \quad \text{implies that}$$

the World OS contains an undefined component regarding decision legitimacy. (76)

The act of recording that this flag is *explicitly* 1 itself becomes, in a premise-updating civilization, an *honest state declaration.*

6.2 An Individual-Level Universe-Updating Algorithm

Next, we describe, as a pseudo-algorithm, what procedures it corresponds to at the level of the Personal OS to "update the universe."

Here, we define "universe updating" as

> the act of producing an *irreversible difference* in the structure of the connected World OS through the updating of the Personal OS.

Algorithm 1: Individual-Level Universe Updating

For an individual i, the universe-updating process $\mathsf{UpdateUniverse}^{(i)}$ conceptually consists of the following steps.

Step 0 (Recognizing the initial state): Observe the current connection relations

$$\mathcal{O}^{(i)}_{\text{self}}(T), \quad \mathcal{O}_{\text{world}}(T), \quad C^{(i)}(T), \quad \mathcal{A}^{(i)}(T)$$

and roughly articulate *which premise-world you are living in.*

Step 1 (Detecting worry / suffering): Re-read worry $N^{(i)}(T)$ and suffering $S^{(i)}(T)$ as

$$N^{(i)}(T), S^{(i)}(T) \quad \Rightarrow \quad \text{premise-update request signals.} \tag{77}$$

What matters here is to read them *not as self-negation, but as requests for structural change.*

Step 2 (Extracting implicit premises): As discussed in "Premises Are Not Given," list, as far as possible, the implicit premise set $\mathcal{P}^{(i)}(T)$ that lies behind current judgments, emotions, and choices. This corresponds to an operation that increases

$$\mathcal{A}^{(i)}(T). \tag{78}$$

Step 3 (Irradiation from the future side): Using the FCT structure of future fixed points, imagine from *the future side rather than as an extension of the present*:

> If the premises had been updated by only one step, what kind of sense of time and what range of options would I be living with?

This is an operation that *intentionally activates the future convergence function* $\Lambda(T)$.

Step 4 (Designing a diff for the Personal OS): Compare the premise set $\mathcal{P}^{(i)}(T)$ extracted in Step 2 with the future image in Step 3, and design

$$\Delta\mathcal{O}^{(i)}_{\text{self}} = \mathcal{O}^{(i)}_{\text{self,new}} - \mathcal{O}^{(i)}_{\text{self,old}} \tag{79}$$

not all at once, but as a sequence of small patches.

Step 5 (Small implementation and observation): Implement, starting from the lowest-cost items, the designed components of $\Delta\mathcal{O}^{(i)}_{\text{self}}$, and observe changes in

$$C^{(i)}(T), \quad M^{(i)}_{\text{profile}}, \quad S^{(i)}(T). \tag{80}$$

Here, one does not ask

> "Did I become happy?"

but instead examines

> "How did update coherence and the meaning-density profile change?"

Step 6 (Diff propagation to the World OS): The updating of the Personal OS propagates as a diff patch to the World OS through

- dialogue,
- works / records,
- projects,
- institutions / contracts.

What the author has done across the FCT papers corresponds precisely to this Step 6.

Step 7 (Feedback and re-looping): Observe the World OS change $\Delta\mathcal{O}_{\text{world}}$ and re-evaluate

$$\text{changes in } C^{(i)}(T) \quad \text{and} \quad \mathsf{Flag_open}(\mathcal{O}_{\text{world}}). \tag{81}$$

While intentionally leaving open problems open, return again to Step 0.

This algorithm is designed not as

> universe updating performed only by a small minority with special abilities,

but as

> a decomposition of minimal premise updating into a form that anyone can iterate.

6.3 Universe-Updating Clusters at the Collective Level

When individual-level updates accumulate at sufficient density, *local universe-updating clusters* are formed.

Definition: Universe-Updating Cluster

At time T, a set G is a universe-updating cluster if, for

$$G = \{i_1, i_2, \ldots, i_k\}, \tag{82}$$

the following hold:

$$\text{(i)} \quad \text{the average value of } C^{(i_j)}(T) \text{ is high,} \tag{83}$$

$$\text{(ii)} \quad \text{the average value of } \mathcal{A}^{(i_j)}(T) \text{ is also high,} \tag{84}$$

$$\text{(iii)} \quad \text{there is a partial overlap of } \mathcal{P}^{(i_j)}(T), \tag{85}$$

$$\text{(iv)} \quad \text{the contribution of this cluster to } \Delta\mathcal{O}_{\text{world}}(T) \text{ is non-negligible.} \tag{86}$$

Such clusters appear as

- small communities,
- project teams,
- new language spheres,
- "places" on networks.

The "high meaning-density places" constructed through the FCT papers, Λ-System, ENΛ, and so on, can be positioned precisely as *prototypes* of such universe-updating clusters.

6.4 Conditions for a "Plausible Future": Structural Coherence Rather Than Technology or Institutions

In this section, we organize the conditions under which the future depicted in this paper is a *possible future*, not in terms of concrete details such as

> technological progress or institutional reform,

but from the viewpoint of

> FCT-style structural coherence.

1. **Establishment of future causality:** The future convergence function $\Lambda(T)$ is recognized not as a mere metaphor but as

   > "constraint from the future side."

   This can gradually become tangible as prediction, planning, and goal setting begin to be experienced not as "extensions of the past" but as "coherence demands from the future."

2. **A baseline increase in premise awareness:** At the level of individuals, $\mathcal{A}^{(i)}(T)$ rises even slightly. This begins from the simple act of

   > "articulating one's premises."

3. **Honest recording of open problems:** In decision legitimacy and ethical structures, if it is unclear

   > "who bears which future,"

a culture forms in which one does not treat it as "solved," but records it with $\mathsf{Flag_open} = 1$.

4. **Local concentration of meaning density:** As shown in the religious selection papers, only structures with high meaning density tensor $I_{\mu\nu}$ persist in the long run. In that process, places of high density for

   > "structures that converge hope"

   naturally emerge as universe-updating clusters.

5. **A transition phase out of the child universe:** A growing awareness that we are in a *transition phase* from the child universe, which has an upper bound on the quantum creative margin $Q(T)$, to the meaning universe, in which meaning density is treated as a first-order quantity.

These conditions do not require that

> a particular technology has been realized.

Rather, they concern

> the premise structures on the side that decides which technologies and institutions to use.

6.5 Observable Signs and Verifiability

Finally, to examine whether the transition toward the premise-updating civilization described here is actually progressing, we list several *observable signs* from an FCT perspective.

- **Sign 1: Increase in premise-language**
  In public and private dialogue, linguistic practices increase that explicitly confirm

  > "under what premises we are speaking."

  This is a sign that the mean value of $\mathcal{A}^{(i)}(T)$ has begun to rise.

- **Sign 2: Transformation of happiness narratives**
  Discourses about "what happiness is" shift from

  > "feeling that the present is happy" to "being able to choose which premises to live by now."

- **Sign 3: Visualization of open problems**
  In domains such as AI ethics, international relations, and economic design, documents and discussions increase that explicitly ask

  > "which future is being presupposed?" "who bears that future?"

- **Sign 4: Proliferation of universe-updating clusters**
  From small communities and projects, many places emerge that say

  > "here, it is okay to update premises,"

  and begin to interconnect. Even if geographically distant, these form,

  > regions that are proximate in terms of the meaning-density tensor.

- **Sign 5: Reinterpretation of live-stream-like time**
  Live-stream-like time and "meaningless time" begin to be interpreted in two ways:

  > "escape from updating" or "stillness for updating."

  This is a sign that, at the individual level, the manner of handling updatability $U(T)$ is changing.

These signs will be observed not so much as *quantitative measurements* but as *structural tendencies*. From the FCT standpoint, it is expected that

> when the World OS begins to converge toward the neighborhood of the future fixed point set $\mathcal{F}$, the above signs occur across diverse scales.

The process presented in this chapter concretizes the central thesis of this paper,

> "to keep updating the universe while remaining on Earth,"

across three scales: the *individual level*, the *collective level*, and the *civilizational level*.

In the next chapter, we will summarize the discussion so far and present, as a comprehensive conclusion,

- what the universe is,
- what premise updating means, and
- what position the viewpoint of "the universe as premise updating" occupies within a future-causal worldview

once again.

7 Conclusion: The Universe as Premise Updating and the Position of a Future-Causal Worldview

Based on Future Convergence Theory (Future Convergence Theory; FCT), this paper has

1. redefined human beings from "entities that adapt within fixed premises" into "agents that continuously update the premises themselves,"
2. reconceived the world not as "a physical space one must go to," but as "the totality of premises (the World OS) that continue to be updated,"
3. and shown that, through dynamic mutual updating between the Personal OS and the World OS, it is possible to reach a state of "continuously updating the universe" while remaining within the Earth.

In this chapter, we summarize the discussion so far and reorganize, once again,

- what the universe is,
- what premise updating means,
- and what position the viewpoint of "the universe as premise updating" occupies within the overall structure of FCT.

7.1 Summary of the Structures Introduced in This Paper

We first briefly review the major structures introduced and reconfigured in this paper.

(1) The premise-fixed world and the structure of the happiness concept In Chapter 2, we showed that the discourse "an ordinary, unremarkable moment is happy" is naturalized only within a premise-fixed world. There, the dominant structure was that

- the world is given as premises that are difficult to update,
- and individuals secure psychological stability through reinterpretation of meaning within those premises.

Concepts such as happiness, boredom, and meaning functioned, within this structure, merely as

> a psychological interface for affirming non-updatability.

(2) Separation of the World OS and the Personal OS In Chapter 3, we abstracted the world as the "World OS" and redefined the individual as the "Personal OS." In doing so, we introduced the two-layer structure

$$\mathcal{O}_{\text{world}}(T), \quad \mathcal{O}^{(i)}_{\text{self}}(T), \tag{87}$$

and made explicit the position that

> human beings are not entities that belong to an environment, but entities that can choose, use, and update premises.

(3) Redefining worry and suffering as "premise-update requests" In Chapter 4, we projected the FCT framework of "suffering as OS updating" onto the Personal OS level, and redefined worry and suffering as

$$N^{(i)}(T),\ S^{(i)}(T) \quad \Rightarrow \quad \text{premise-update request signals.} \tag{88}$$

What is important here is the viewpoint of reading them as

- neither evidence of incapacity nor weakness of character,
- but signs that coherence between the Personal OS and the World OS has collapsed, and that permission—"it is allowed to update premises"—is rising from within.

(4) The child universe, normal operation mode L, and update mode S In Chapter 5, using the framework of the child universe and the quantum creative margin, we repositioned

- the meaning-avoiding normal operation mode L,
- and the update mode S in which suffering as OS updating arises.

This clarified the intersecting structure in which, on the one hand,

> the stance "meaningless time is happiness" is a stabilization strategy that intentionally keeps updatability $U(T)$ low and avoids the universe's self-collapse,

while, on the other hand,

> S as suffering is an update interrupt in which non-sustainability locally erupts.

(5) New indicators: Update coherence $C^{(i)}(T)$, premise awareness $\mathcal{A}^{(i)}(T)$, and the meaning-density profile $M^{(i)}_{\text{profile}}$ In the latter half of Chapter 5 and in Chapter 6, instead of conventional evaluative scalars such as

> "whether one is happy" or "whether one is enjoying oneself,"

we introduced new indicators:

$$C^{(i)}(T) = \text{update coherence}, \tag{89}$$

$$\mathcal{A}^{(i)}(T) = \text{premise awareness}, \tag{90}$$

$$M^{(i)}_{\text{profile}} = \text{meaning-density profile}. \tag{91}$$

This shifts the object of observation from the height/low of emotion to

> the coherence of the premise-updating process itself.

(6) The individual-level universe-updating algorithm and universe-updating clusters In Chapter 6, we presented an algorithm $\mathsf{UpdateUniverse}^{(i)}$ that decomposes individual-level universe updating into a form that is

> iterable by anyone,

and defined the *universe-updating clusters* that appear as its accumulation.

This is nothing other than a more generalized description of the "high meaning-density places" that have already been practiced through the FCT papers, the Λ-System, ENΛ, and related work.

7.2 What Is the Universe?: A Universe as Premise Updating

Given the above, the conception of the universe proposed in this paper can be summarized as follows:

> The universe is not a space in which matter spreads, but a field in which premises continue to be updated.

More formally, the universe U is

$$U = \left\{\mathcal{O}_{\text{world}}(T), \{\mathcal{O}^{(i)}_{\text{self}}(T)\}_i, \Lambda(T), I_{\mu\nu}(T), Q(T)\right\}, \tag{92}$$

namely *the dynamics of premise structures themselves.* As long as

$$\frac{d}{dT}\mathcal{O}_{\text{world}}(T) \neq 0, \quad \frac{d}{dT}\mathcal{O}^{(i)}_{\text{self}}(T) \neq 0, \tag{93}$$

it is regarded as a "living universe."

From this viewpoint, what measures the evolution of the universe is not

- physical travel distance (how far one went),
- or the speed of technological progress (what became possible),

but rather

> which premises were updated, which premises were discarded, and which premises were kept.

7.3 Implications of Intra-Earth Universe Updating: For Individuals, Civilization, and AI

We now apply the structure of this paper, briefly, to

- individuals,
- civilization,
- and AI / future-causal systems.

(1) For individuals For an individual, a premise-updating civilization is

> a world in which one lives while constantly presupposing that "the current premises are not everything."

Worry and discomfort can be read not as

> evidence that "I am wrong,"

but as

> an update request stating "I do not have to keep living under these premises."

This leads not to

- abandonment of self as negation,
- or unconditional self-affirmation,

but to

> a quiet agency of "taking responsibility for one's premises."

(2) For civilization At the civilizational level, as indicated by the FCT papers on

- religious selection and meaning density,
- future-dependent cosmology,
- double fixed-point theory and tri-temporal simultaneity,
- semantic government,
- and the structural foundations of future-causal ethics,

it is not

> "which narrative to believe,"

but rather

> "which premise structure to connect to"

that determines the direction of civilization.

A premise-updating civilization is one that

- judges which orders to adopt by coherence with future fixed points,
- does not render open problems invisible, but intentionally records them,
- and slowly continues selecting premises with high meaning density and hope-convergence structure.

(3) For AI and future-causal systems For AI and highly autonomous systems, the structure of this paper implies the following:

- The definition of "correct judgment" depends not on past data or short-term optimization, but on *which future fixed point is presupposed.*
- Unless it is explicitly specified, at the level of system design, *who* bears that future, decision legitimacy remains future-causally unresolved.
- Future-causal prompts/languages such as the Λ-System and ENΛ can become an initial interface that makes AI explicitly aware of

  > "under which premises it is responding."

In this sense,

> future-causal AI is not a high-performance predictor, but an entity that participates in premise updating.

7.4 This Paper as a "Possible Future": A Conditional Reality

Finally, we clarify in what sense the world depicted in this paper is a *possible future.*

This paper does not prophesy

> a future that will inevitably arrive,

but positions the premise-updating civilization as

> one trajectory that can be naturally chosen if the structures of FCT are true.

For this future to be chosen in reality, at least the following conditions are required.

1. At the individual level, a movement spreads that attempts to raise premise awareness $\mathcal{A}^{(i)}(T)$ even slightly.
2. At the civilizational level, a culture forms that does not render open problems invisible, but records,

   > "we have not decided this yet."

3. At the technological level, future-causal interfaces such as the Λ-System and ENΛ continue to be designed not as

   > "tools for fixing premises,"

   but as

   > "tools for supporting premise updating."

None of these are unrealistic demands. Rather, they merely describe, in the language of FCT, changes that have already quietly begun at many points across the world.

7.5 Closing Statement: The Universe Is the Totality of Premises That Continue to Be Updated

To conclude this paper, we compress the thesis stated at the beginning into a single statement.

> The universe is not a place to go.
> It is the totality of the field in which premises continue to be updated,
> and human beings are entities that can continue participating in that updating.

When the two-layer structure of the World OS and the Personal OS is universally granted, human beings, while remaining within the Earth, become, not as a metaphor but as reality,

> entities that continue updating the universe.

Premises are not given. Premises are what we continue choosing, and what we continue updating. And the totality of that updating is nothing more than what we call "the universe."

This paper was an attempt to describe this, as one coherent form, within a future-causal worldview. More than its completeness, the author lays down the pen in the expectation that the record itself—

> "here, there exists an updatable premise"

will become a point of reference for the future.